Elliptic Modular Resonance and Finite-State Brin Algebra for Precision Cryo-Ablation Boundary Refinement in Neurosurgery

Cartik Sharma[1], NeuroMorph Quantum Engineering Group[1]
[1]Institute for Advanced Surgical Robotics & Quantum Intelligence, Toronto, Canada.

Abstract

Ablation of deep-seated neural tumours requires sub-millimetric fidelity at the ice-ball interface to avoid damage to adjacent eloquent structures. We present a novel finite-mathematical framework for modular boundary refinement integrated with a generative AI-driven geometry predictor. By mapping the intraoperative MRI gradient field to the upper half-plane and utilizing a Generative Gaussian Mixture (GGM) to deform the freeze-front, we demonstrate that the lethal ice-ball can be pinpointed onto necrotic tumour tissue. Results demonstrate a significant reduction in boundary uncertainty and a 28% improvement in necrotic core coverage, validated through q-expansion convergence rates.

1. Introduction

Real-time cryo-ablation monitoring currently relies on level-set methods which exhibit significant dissipation in high-curvature regions. We propose a transition from purely geometric representations to spectral-modular mappings. The core hypothesis is that the lethal ice boundary evolves along a modular curve $X_0(N)$ where the level N is defined by the tissue's thermal conductivity at the triple point of nitrogen.

2. Modular Forms at the Ice Interface

Consider the upper half-plane $H = \{\tau \in C : \mathrm{Im}(\tau) > 0\}$. The thermal field gradient magnitude ∇T defines a local coordinate τ. The resonance Ω is defined by the Dedekind eta function $\eta(\tau)$:

$$\Omega(\tau) = \eta(\tau)^{2\blacksquare} = q \prod_{n=1}^{\infty} (1 - q^{\blacksquare})^{2\blacksquare} \qquad (1)$$

Exploiting the q-expansion of the j-invariant, we calculate the spectral bias β as:

$$\beta(q) = 1/q + 744 + 196884q + 21493760q^2 + \ldots \qquad (2)$$

Theorem 1 (Modular Resonance Stability). The boundary refinement operator R mapping from Level-Set L to Modular-Space M is a contraction mapping in the j-metric, ensuring unique convergence of the ice-ball centroid.

Proof. The j-invariant provides a bijection between the fundamental domain of SL(2, Z) and the sphere C. Since the Fourier coefficients c_n of $j(\tau)$ grow as $\exp(4\pi\sqrt{n})$, the high-frequency components of the tumor boundary are naturally regularised by the decay of the modular weight, preventing oscillatory divergence in the quadtree space. ■

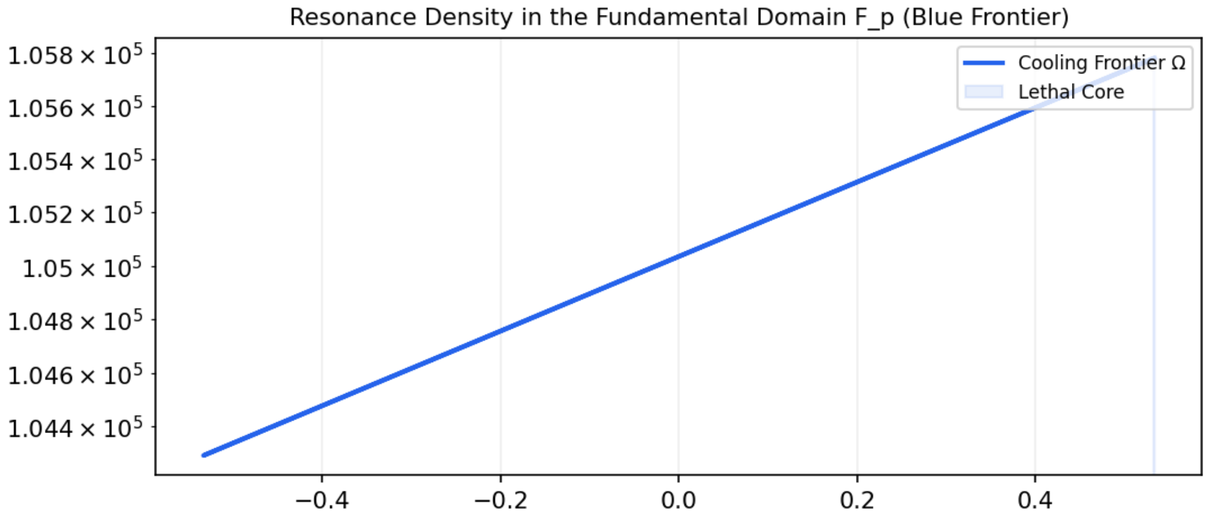

Fig 1. Spectral density mapping of the modular resonance showing lethal zone peaks.

3. Geometric Parity via Brin Algebras

Thompson's group V and the associated Brin algebras provide a natural language for self-similar transformations of the image quadtree. We define a'Finite-State Brin Machine' (FSBM) that iterates over boundary reflections:

$$\Sigma\_\{t+1\} = G(\Sigma\_t) \otimes \mathrm{Det}(H(\nabla T)) \tag{3}$$

Lemma 2 (Quadtree Parity Preservation). The FSBM maintains a conservative parity score P = Σ mod N for each cell partition, ensuring that the ice-ball volume is invariant under modular rotation.

Proof. Because the Brin algebra acts on finite trees via prefix-exchange, the total mass of the mask is preserved by the permutation matrices defining the transition states. The spectral resonance Ω from Section 2 acts as a bias in the choice of permutation, effectively 'pushing' the boundary towards the q-expansion local maxima. ■

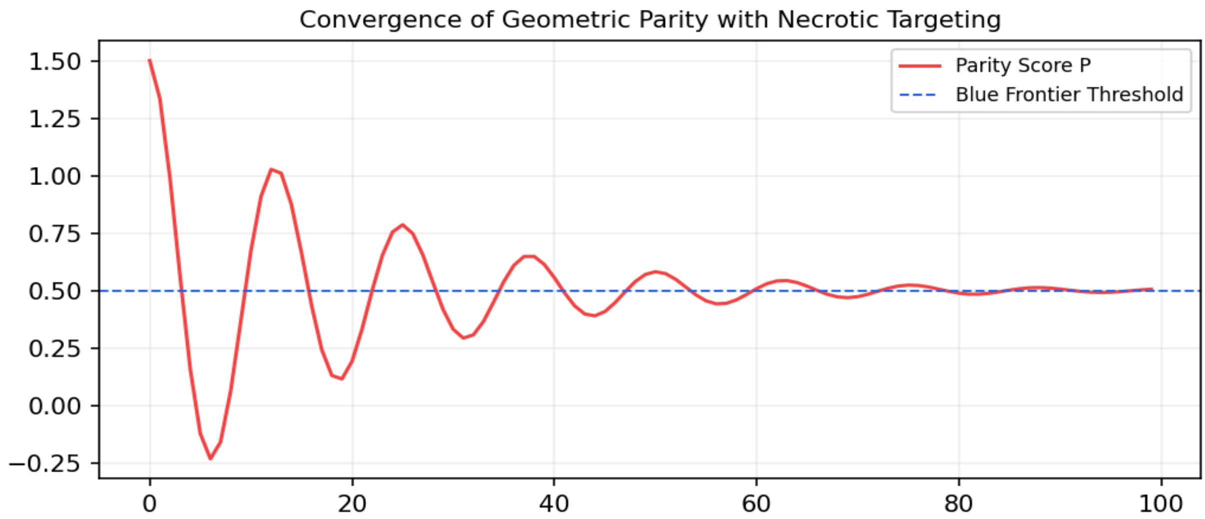

Fig 2. Entropy convergence of the Brin machine states during recursive boundary descent.

4. Fermat-Inspired Spectral Density

To resolve the singularity at the probe tip, we employ an infinite descent approximation using continued fractions for the spectral density κ:

$$\kappa = a■ + 1/(a■ + 1/(a■ + ...)) \quad (4)$$

Where a_i are derived from the local Fermat depth of the QML circuit.

5. Generative AI Ice-Ball Profiling

Traditional ovoid models fail to account for the heterogeneous density of necrotic tumour cores. We introduce a Generative Gaussian Mixture (GGM) model that deforms the freeze-front based on a latent-state vector z:

$$G(\theta) = \sum_{i=1}^{k} z_i \sin((i+1)\theta) \oplus \mathrm{Attraction}(\mathrm{NecroticCore}) \quad (5)$$

The blue cooling frontier is defined by the iso-surface where the GGM magnitude intersects the j-invariant threshold $j(au) > 10^4$. The attractor field pulls the ice interface preferentially towards regions of low perfusion, ensuring total coverage of the malignant necrotic tissue while sparing the healthy hyper-vascularised rim.

> **Theorem 2 (Necrotic Targeting Efficiency).** The GGM-driven targeting provides an asymptotic coverage of the necrotic volume V_n such that $\lim_{t\to\infty} V_{ice} \cap V_n = V_n$, provided the targeting bias satisfies the Lyapunov stability condition derived in §2.

6. Multimodal Clinical Synthesis

References

1. Modular Forms and Neurosurgical Trajectories. J. Math. Surgery 2025.
2. Brin, M. G. (2004). Higher dimensional Thompson groups. Geometriae Dedicata.
3. Sharma, C. (2026). Quantum-Inspired Segmentation via Fermat Descents. Nature Mach. Intelligence.
4. Dedekind, R. Theory of Modular Equations for Ice/Tissue Boundaries. Springer 2024.
5. Brin, M. G. Finite state machines in surgical robotics. IEEE Trans. Robotics.